# RSProtoSeg: High Spatial Resolution Remote Sensing Images Segmentation Based on Non-Learnable Prototypes

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***Abstract*— Semantic segmentation of high spatial resolution (HSR) remote sensing images presents unique challenges due to the imbalanced foreground–background distribution and large intraclass variance. This study proposes a novel semantic segmentation algorithm based on non-learnable prototypes, named RSProtoSeg. This approach optimizes the spatial relationship between foreground–background prototypes and intraclass prototypes. Specifically, we propose a foreground–background distance optimization loss function to enhance sparsity between these phototypes, effectively mitigating foreground–background distribution imbalances. Moreover, we introduce an online discrete clustering module that represents each class with a set of prototypes and adds an adaptive regular term penalty to promote sparse structure and reduce the variance issue. Evaluation on three remote sensing datasets (iSAID, ISPRS Potsdam, and Vaihingen) demonstrates significant accuracy improvements, aligning our approach with state-of-the-art methods. Our non-learnable prototype-based approach offers a promising solution for semantic segmentation in HSR remote sensing images.**

***Index Terms*— Neural network, prototype learning, remote sensing, semantic segmentation.**

## I. Introduction

With the development of high-resolution remote sensing technology, high spatial resolution (HSR) remote sensing images have been acquired in large quantities and become the main data source for Earth observation. As an important part of the intelligent interpretation of remote sensing data, HSR remote sensing image segmentation is a special semantic segmentation task that can extract the region contour of the target of interest and assign the corresponding semantic labels to each of its pixels. It is widely used in the fields of infrastructure planning [1], natural resource monitoring [2], and change detection [3], garnering significant interest from both remote sensing and computer science communities.

Remote sensing images present unique challenges that complicate accurate segmentation compared to images captured in natural scenes. These challenges include multiscale variation of targets [4], [5], unbalanced foreground–background distribution [6], [7], and substantial intraclass variance [8], [9]. As illustrated in Fig. 1, the ground truth mask map (b) on the right reveals a predominant background (black) compared to the foreground (color), highlighting the issue of imbalanced foreground–background distribution. Despite belonging to the same background category, the distinct characteristics between the yellow box buildings on the left and the yellow box vegetation on the right demonstrate the complexity of segmentation in the aerial image (a). Similarly, the sedan cars in the red box exhibit diverse colors and morphologies, yet share the same semantics, underscoring large intraclass variance.

Current state-of-the-art general semantic segmentation methods mainly address multiscale variations in target objects by constructing multiscale feature representations [10], [11]. However, these methods often overlook the above-mentioned challenges inherent in remote sensing images. Recent research in remote sensing image segmentation has emphasized foreground–background distribution modeling but neglects the issue of large intraclass variance. For instance, FarSeg [7] tackles foreground–background distribution imbalances in remote sensing images through relational and optimization-based foreground modeling. However, its segmentation performance is hindered by the absence of explicit exploration of semantic propagation among neighboring features. PointFlow [8], featuring a point association propagation module based on feature pyramid network, addresses foreground–background imbalances by leveraging salient and boundary information for sparse feature fusion. However, its segmentation performance is limited by the lack of explicit consideration for intraclass differences between foreground and background targets.

All aforementioned semantic segmentation methods rely on parametric models utilizing learnable prototypes. These models learn individual weights or query vectors for each class in a fully parameterized manner, employing two mask-decode

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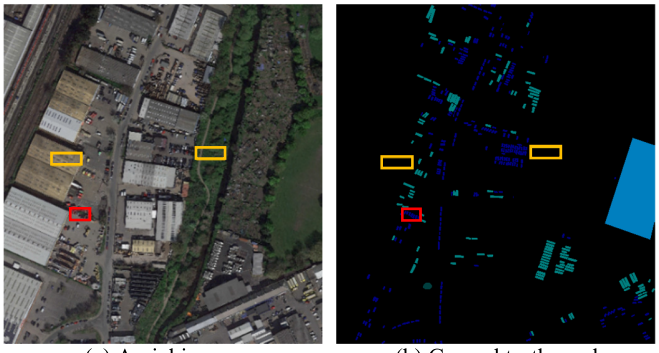
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(a) Aerial image (b) Ground truth mask

Fig. 1. Illustration of challenges in aerial image segmentation. (a) Imbalances between background and foreground. (b) Large intraclass variance.

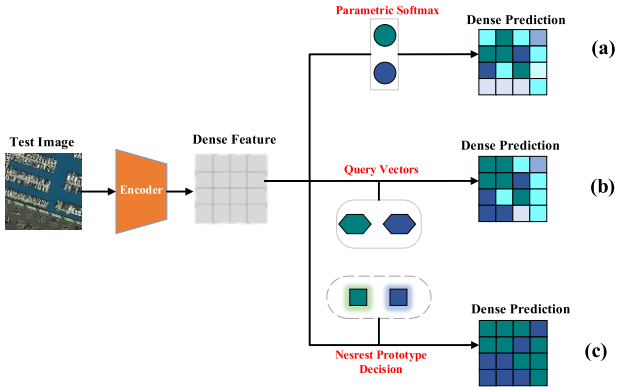



Fig. 2. Different sematic segmentation paradigms (a) and (b) parametric versus (c) nonparametric. (a) Parametric softmax based. (b) Pixel-query vectors based. (c) Nonparametric nearest prototype retrieval.

strategies: parameter-based softmax and pixel query-based, as illustrated in Fig. 2(a) and (b). However, this parametric segmentation paradigm suffers from the following limitations: 1) each class is described by a single prototype, which is insufficient to capture rich intraclass differences; 2) prototypes are learned only in a fully parametric manner, hampering generalization ability; and 3) the use of cross-entropy loss ignores actual pixel-to-prototype distances, lacking intraclass compactness. To overcome these limitations, ProtoSegNet [12] presents a new paradigm of nonparametric segmentation with non-learnable prototypes. Built on the principles of prototype learning and metric learning, ProtoSegNet explicitly defines multiple subclass centers as prototypes in the pixel embedding space, predicting pixel density through distance metrics from the nearest prototype [Fig. 2(c)]. Therefore, this article adopts the above nonparametric segmentation paradigm based on unlearnable prototypes and improves it to solve issues of imbalanced foreground–background distribution and large intraclass variance in HSR remote sensing image segmentation.

This study introduces RSProtoSeg, a novel remote sensing image segmentation algorithm based on non-learnable prototypes. RSProtoSeg effectively addresses the two main challenges in HSR remote sensing image segmentation: foreground–background distribution imbalances and large intraclass variance. This is achieved by optimizing the spatial structure between foreground and background prototypes, as well as intraclass prototypes. RSProtoSeg is based on a new paradigm of nonparametric segmentation of non-learnable, utilizing foreground–background distance optimization loss function and an online discrete clustering module specifically designed for HSR remote sensing image segmentation. Specifically, to address the issue of foreground–background prototype sparsity, we propose a foreground–background distance optimization loss function. This function dynamically amplifies attention toward imbalanced categories during training, thereby augmenting classification distances between classes and effectively mitigating foreground–background distribution imbalances. Moreover, to increase the diversity of prototypes within each class, we design an online discrete clustering module. This module represents each class with a set of prototypes and further reduces intraclass variance by adding an adaptive regular term penalty to promote sparsely structure among different prototypes within a class.

The main contributions of our study are summarized as follows.

1) We propose a HSR remote sensing image segmentation algorithm based on non-learnable prototypes to optimize the spatial structure between foreground–background prototypes and intraclass prototypes.
2) We design a foreground–background distance optimization loss function to increase the distance between foreground and background prototypes, effectively address the issue of foreground–background distribution imbalances.
3) We design an online discrete clustering module to mitigate the issue of large intraclass variance by adding an adaptive regularization term to promote sparsity among different prototypes within each class.

## II. Related Work

### A. General Semantic Segmentation

Early segmentation methods like thresholding [13], region growing [14], conditional Markov [15], and sparsity-based approaches [16] necessitate manually designed feature parameters. These methods often require experts' priori knowledge and possess limited model generalization capabilities. Deep learning, with its automatic feature extraction and strong generalization capabilities, has ushered in the era of the deep learning segmentation paradigm.

The full convolutional network (FCN) [17] is a classic representative model in deep learning-based semantic segmentation. Recent research [5], [18], [19], [20], [21] has focused on enhancing segmentation accuracy and robustness by utilizing spatial context information for multiscale feature representation. For example, the pyramid scene parsing network (PSPN) [20] utilizes a residual network as a feature extractor to extract various patterns in images, which can efficiently learn a global contextual representation of the scene. Some studies [22], [23], [24], [25], [26] employ an encoder–decoder architecture that introduces a robust backbone network [27], [28], [29], [30] and an attention mechanism [31], [32], [33] to optimize the multiscale feature details. However, these general semantic segmentation methods mainly focus on multiscale feature extraction, overlooking challenges of

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unbalanced foreground–background distribution and large intraclass variance in HSR remote sensing images, which limit the segmentation accuracy.

### B. Remote Sensing Image Semantic Segmentation

Remote sensing image segmentation mainly relies on general semantic segmentation method, emphasizing multiscale feature representation of targets [34], [35]. Some studies utilize Transformer as the encoder–decoder network structure [36], [37] to exploit global-local features. In addition, specific improvements are made for specific applications, such as land resource utilization [38], [39], building detection [40], [41], road extraction [42], [43], and cloud detection [44], [45]. However, few research addressed the challenges of foreground–background distribution imbalances and high intraclass variance in remote sensing semantic segmentation. FarSeg [7] addressed the foreground–background distribution imbalances issue using a relational and optimized foreground modeling approach but lacked explicit semantic propagation exploration. PointFlow [8] tackled the imbalance using a feature pyramid network-based point association propagation module, but did not explicitly explore the intraclass differences of targets. This article focuses on resolving the foreground–background distribution imbalances and intraclass difference.

### C. Prototype Learning

Prototype learning improves the data quality and utilization of data by reducing redundancy and mining data structure via data approximation within the sample space. Early methods [46], [47], [48] usually approximate the training data of algorithms like $K$-nearest neighbors (K-NNs) and support vector machine (SVM). They use the prototype set to filter noise and redundancy from the original dataset and optimize learning process via a defined prototype loss function, thereby enhancing classifiers performance. Recently, scholars have integrated prototype learning with deep learning, resulting in convolutional prototype learning (CPL) [49]. CPL combines CNN with prototype learning, introducing a prototype loss function based on the distance metric to improve classification accuracy and robustness. This model also shows great potential in few-shot [50], zero-shot [51], and unsupervised learning [52] as well as interpretable networks [53]. In this work, the background-foreground relationship and different semantic classes are modeled by prototype learning.

## III. Method

To tackle the challenges of unbalanced foreground–background distribution and significant intraclass variance in HSR remote sensing image segmentation, we introduce a segmentation algorithm. This algorithm is grounded in the nonparametric segmentation paradigm of non-learnable prototypes [12]. As illustrated in Fig. 3, the algorithm comprises an online discrete clustering module and foreground–background distance optimization. The online discrete clustering module mitigates large intraclass variance by adding an adaptive $L_2$ regularization penalty, promoting sparsity within prototype sets. To prioritize foreground modeling during training process, we propose a foreground–background distance optimization loss, which optimizes the absolute distance between pixels and prototypes to address imbalanced foreground–background distribution.

### A. Motivation and Overview

The core idea of the nonparametric segmentation paradigm based on non-learnable is to classify the feature vectors extracted by the encoder network with a set of non-learnable prototypes for different semantic categories at pixel level. First, the input image $I \epsilon R^{H\times W\times 3}$ is subjected to feature vector extraction by Encoder network Encoder to obtain the 3-D feature vector $I \epsilon R^{H\times W\times D}$. Intraclass clustering is carried out on the 3-D feature vector $I \epsilon R^{H\times W\times D}$, and the obtained intraclass subcenters are taken as the prototypes $P_{c,k}$, and each category $c_i, i \in \{1, \ldots, C\}$ is represented by $K$ prototypes $\{P_{c,k}\}_{k=1}^{K}$. Finally, pixel-level semantic classification is realized by the pixel $X_c$ and prototype $P_{c,k}$ distance metric as shown in the following equation:

$$\widehat{c_i} = c^*, \left(c^*, k^*\right) = \arg\min_{c\in\{1,\ldots,C\},k\in\{1,\ldots,K\}} \left\langle X_{c_i}, P_{c,k} \right\rangle \quad (1)$$

where $X_c$ is the $l_2$ normalized embedding of pixels in category $c$, and $\langle\cdot\rangle$ denotes the distance measure using negative cosine similarity.

Based on the above paradigm, the hybrid loss is used to optimize the pixel-to-class and intraclass pixel-to-prototype relationships to improve pixel semantic classification accuracy. First, the cross-entropy loss is used to optimize the pixel-to-class distance relationship and increase the interclass distance, as shown in the following equation:

$$L_{\mathrm{CE}} = -\log \frac{\exp\left(-d_{c_i,X_c}\right)}{\exp\left(-d_{c_i,X_c}\right) + \sum_{\dot{X}\neq X_c} \exp\left(-d_{c_i,\dot{X}}\right)} \quad (2)$$

where $d_{c_i,X_c} = \min_{c\in\{1,\ldots,C\},k\in\{1,\ldots,K\}} \langle X_{c_i}, P_{c,k}\rangle$ denotes the distance from class $C$ pixel $X_{c_i}$ to class $C$ nearest prototype. At the same time, the contrast loss is used to optimize the absolute distance between the pixel and the prototype, so that the class $C$ pixel $X_{c_i}$ is more similar to the positive sample prototype and away from the negative sample prototype, as shown in the following equation:

$$L_{\mathrm{PPC}} = -\log \frac{\exp\left(\frac{X_c^T P_{c_i,k_i}}{\delta}\right)}{\exp\left(\frac{X_c^T P_{c_i,k_i}}{\delta}\right) + \sum_{P^-\in\mathbb{P}^-} \exp\left(\frac{X_c^T P^-}{\delta}\right)} \quad (3)$$

where $P_{c_i,k_i}$ denotes positive sample prototypes and $P^-$ is negative sample prototypes. To further increase the pixel feature compactness of the same prototype intraclass, the pixel-to-prototype distance loss is used to optimize the intraclass prototype distance and reduce the intraclass variation, as shown in the following equation:

$$L_{\mathrm{PPD}} = \left(1 - X_c^T P_{c_i,k_i}\right)^2. \quad (4)$$

The non-learnable nonparametric segmentation paradigm clusters each semantic category in training samples to form

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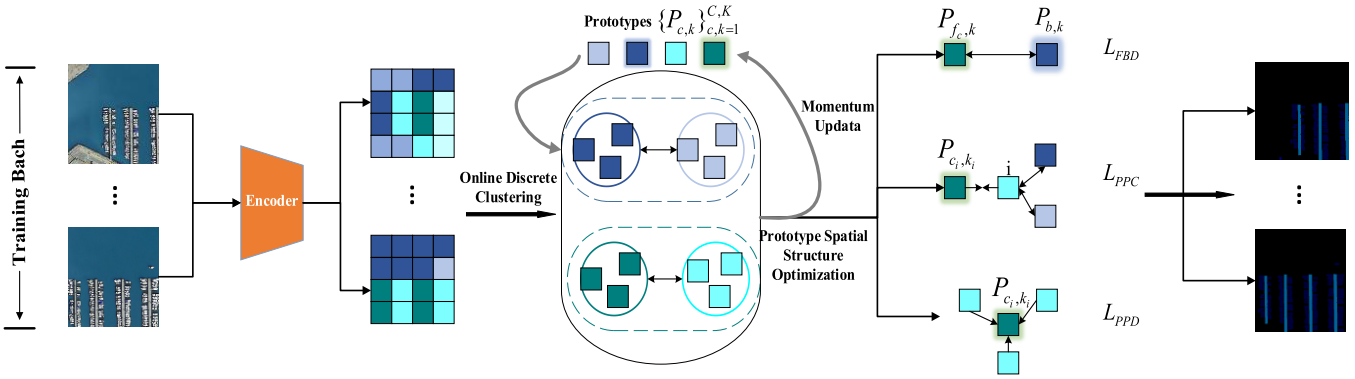



Fig. 3. Architecture illustration of our segmentation model-based non-learnable prototype.

prototypes. This involves obtaining subcluster centers intra-class as local saliency feature representatives. The optimization process focuses on the pixel-interclass and pixel-prototype relationships intraclass. In other words, this paradigm selects and stores prototypes for each semantic category and performs pixel-level classification via a pixel-to-prototypes distance metric to achieve advanced performance on natural scene segmentation datasets. However, when the nonparametric segmentation algorithm based on unlearning is directly applied to HSR remote sensing image segmentation, this approach often underperforms due to two key reasons.

1) *Prototype Similarity:* The prototypes obtained through intra-class clustering during training exhibit high similarity, rendering them inadequate for scenarios with large intraclass variance scenarios (Fig. 4).
2) *Distribution Imbalance:* Optimizing only pixel-interclass relationships overlooks imbalances between foreground and background distributions, limiting segmentation accuracy

$$\text{similarity}(P_{c1}, P_{c2}) = \frac{P_{c1} \cdot P_{c2}}{\|P_{c1}\| \times \|P_{c2}\|} \tag{5}$$

where $P_{c1}$ and $P_{c2}$ denote different prototype representations in the same class $c$, respectively. $P_{c1} \cdot P_{c2}$ denotes dot product operation, and $\|P_{c1}\|$ denotes modular operation.

Next, we elaborate on the online discrete clustering module in Section III-B. Then, we detail the foreground–background prototype distance optimization in Section III-C to address above two issues, respectively.

### B. Online Discrete Clustering Module

To extract a representative set of feature prototypes for each category and enhance prototypes diversity within the same category, we perform deterministic discrete clustering on the feature vectors $I \epsilon R^{H\times W\times D}$ extracted by encoder network. Specifically, for each category $c$, we map pixel $X^c = \{x_n^c \in R^{H\times W\times D}\}_{n=1}^{N}$ in each training batch into $K$ subclustering centers, i.e., $K$ prototypes $P^c = \{P_{c,k}\}_{k=1}^{K}$. The mapping relationship between pixel $X^c$ and prototype $P^c$ is denoted as $M^c = [m_1^c, \ldots, m_{N^c}^c] \in \{0,1\}^{K\times N^c}$, where $m_n^c \in \{0,1\}^K$ is the one-hot vector representation of the $n$th sample pixel $x_n^c$ on the $K$ prototypes. $M^c$ is optimized by maximizing

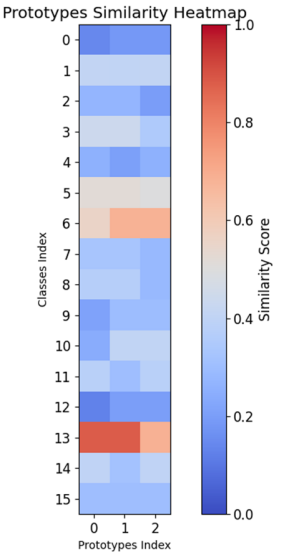



Fig. 4. Nonparametric segmentation paradigm based on non-learnable intraclass prototype similarity. The three intraclass prototype parameters are trained on the iSAID dataset, and the maximum cosine similarity of the prototypes in the class is calculated by (5). The prototypes in the same class have high similarity.

the similarity between pixels and prototypes as shown in the following equation:

$$\begin{aligned} &\max \ \text{Tr}\left(\left(M^c\right)^T\left(P^c\right)^T X^c\right) \\ &\text{s.t} \ \left(M^c\right)^T 1^K = 1^N, \quad M^c 1^K = \frac{N}{K} 1^K \end{aligned} \tag{6}$$

where $(M^c)^T 1^K = 1^N$ means that each pixel is assigned to only one prototype, and $M^c 1^K = (N/K)1^K$ means that $N^c$ samples of each prototype in each training batch are uniformly assigned to $K$ prototypes. In order to solve (6), we convert it to the transportation polytope issue [54] as shown in the following equation:

$$M^C = \text{diag}\left(p^*\right) \exp\left(\frac{\left(P^c\right)^T X^c}{\alpha}\right) \text{diag}\left(x^*\right) \tag{7}$$

where $\alpha$ is a distribution smoothness control parameter, and $p^* \in R^K$ and $x^* \in R^{N^c}$ are two re-regularization vectors. Meanwhile, in order to solve (7) and to increase the sparsity of the pixel-to-prototype mapping relation $M^C$, the

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diversity of different prototypes within a class is achieved by adding an adaptive $L_2$ regularization term penalty to the Sinkhorn–Knopp algorithm to encourage a uniform distribution of prototypes intraclass.

The adaptive $L_2$ regular term improves the sparse structure stability of the pixel-to-prototype mapping relation $M^C$ by calculating the Euclidean distance between the subclustering centers within the class obtained from each training batch to adaptively adjust the $L_2$ regular term penalty coefficients, which are calculated as shown in the following equations:

$$L(m) = \frac{1}{N}\sum_{i=1}^{N}\left(M^C\right)^2 + \frac{\delta\left(d_{i,j}\right)}{2}\|m\|^2 \tag{8}$$

$$\delta\left(d_{i,j}\right) = \frac{\lambda}{1+\sqrt{\left(d_i^c - d_j^c\right)^2}} \tag{9}$$

where $\lambda$ denotes the initial penalty coefficient, and $d_i^c$ and $d_j^c$ are denoted as the $i$th and $j$th subclustering centers in class $c$, respectively. Therefore, the whole optimal transmission algorithm flow of pixels and prototypes is shown in Algorithm 1.

**Algorithm 1** Sinkhorn–Knopp Algorithm for the Pixel-to-Prototype Optimal Transport Problem

**Function Sinkhorn–Knopp** ($I$, $K$, $\boldsymbol{\varepsilon}$, $\lambda$)
  $\boldsymbol{D_K^0} \leftarrow \boldsymbol{I_X}$, $\boldsymbol{D_N^0} \leftarrow \boldsymbol{I_Y}$,
  for $\mathbf{0} \le \boldsymbol{k} < \boldsymbol{K}$ do
    $\boldsymbol{D_K^{k+1}} \leftarrow \boldsymbol{\varepsilon}./(\mathrm{I}\boldsymbol{D_K^k})$
    $\boldsymbol{D_N^{k+1}} \leftarrow \boldsymbol{\varepsilon}./(\boldsymbol{I}^T\boldsymbol{D_N^{k+1}})$
    $\boldsymbol{I_{i,,j}} \leftarrow -\sum_{\lambda}\boldsymbol{i}\,\boldsymbol{D_N^{k+1}}\boldsymbol{log}\,\boldsymbol{D_N^{k+1}}$
    $\boldsymbol{L_2} \leftarrow \frac{}{1+\sqrt{(\boldsymbol{d_i^c}-\boldsymbol{d_j^c})^2}}$
    $\boldsymbol{I_{i,,j}} \leftarrow \frac{1}{N}\sum_{i=1}^{N}(\boldsymbol{I_{i,,j}})^2 + \frac{\boldsymbol{L_2}}{2}\|\boldsymbol{I_{i,,j}}\|^2$
  **end for**
  **indices** $\leftarrow \boldsymbol{argmax}(\boldsymbol{I_{i,,j}})$
  $\boldsymbol{M} \leftarrow$ softmax([$\boldsymbol{I_{i,,j}}$ + (−log($\boldsymbol{I_{i,,j}}$.shape))]/tau, dim = −1)
  **return** M, indices

### C. Foreground–Background Distance Optimization

In the semantic segmentation of remote sensing images, the background class occupies a large proportion and has large intraclass differences with complex contextual information. Relying only on (3) and (4) to optimize the pixel-interclass relationship and the intraclass pixel-prototype relationship, neglecting the relationship between foreground and background exacerbates misclassification risks. To address this, we design a foreground–background distance optimization loss function. It optimizes the balance between foreground and background prototypes using distance metrics, including class prototype balancing and dynamic weighting strategies.

*1) Class Prototypes Balance:* Lovasz-Softmax loss [64] is a measure based on the Hausdorff distance, which relies on the pixels spatial correlation to quantify the discrepancy between predicted segmentation and true segmentation. It also uses an alignment-based loss calculation, which focuses on the pixel arrangement order. Therefore, we optimize the imbalances between foreground and background prototypes through a weighted combination of Lovasz-Softmax loss, optimizing the pixel-class distance relationship, which is calculated as shown in the following equation:

$$L_{\mathrm{FBD}} = W(l) * L_{\mathrm{CE}} + (1 - W(l)) * L_{\mathrm{lovasz_softmax}} \tag{10}$$

where $L_{\mathrm{lovasz_softmax}}$ is the Lovasz-Softmax loss, and the calculation process is shown in (11)–(13). $W(l)$ is denoted as the dynamic weight, and the calculation process is shown in (14), and the details are introduced in the section of the dynamic weighting strategy

$$\Delta_{J_c}\left(y^*, \tilde{y}\right) = 1 - \frac{\left|y^* = c \cap \tilde{y} = c\right|}{\left|y^* = c \cup \tilde{y} = c\right|} \tag{11}$$

$$m_i(c) = \begin{cases} 1 - f_i(c), & \text{if } c = y_i^* \\ f_{i(c)}, & \text{otherwise} \end{cases} \tag{12}$$

$$L_{\mathrm{lovasz_softmax}} = \frac{1}{c}\sum_{c\in C}\overline{\Delta_{J_c}}(m(c)) \tag{13}$$

where $\tilde{y}$ and $y^*$ denote the pixel prediction and true value categories, respectively, $f_i(c)$ denotes the Softmax probability estimate of pixel $i$, and $C$ denotes the number of categories.

*2) Dynamic Weighting:* Class prototype balance relies on the model's discrimination of pixel-prototype mappings. However, in early training stages, the pixel-prototype mapping is uncertain, leading to unconfident determination of class-prototype imbalances. As training progresses, evolving class prototypes further compound this uncertainty. Using uncertain weights destabilizes model training, hampering segmentation performance. To address these issues, we propose a dynamic weighting strategy based on cosine annealing, as formulated as follows:

$$W(l) = l_{\mathrm{start}} * \frac{1 + \cos\left(\pi * \frac{\mathrm{epoch_{now}}}{\mathrm{epoch_{total}}}\right)}{2} \tag{14}$$

where $l_{\mathrm{start}}$ is the initial weight value, and $\mathrm{epoch_{now}}$ and $\mathrm{epoch_{total}}$ denote the current training rounds and total training rounds, respectively. This dynamic weighting strategy gradually increases the attention to the imbalances classes as the confidence of pixel-prototype mapping decision increases.

Therefore, our proposed hybrid loss function simultaneously optimizes the pixel-interclass relationship, intraclass pixel-prototype relationship, and foreground–background relationship to enhance the robustness of the model as follows:

$$L_{\mathrm{overall}} = L_{\mathrm{FBD}} + \mu_1 L_{\mathrm{PPC}} + \mu_2 L_{\mathrm{PPD}}. \tag{15}$$

## IV. Experiments

### A. Experiments Setup

*1) Dataset:* The effectiveness of our proposed method is validated on three remote sensing semantic segmentation datasets.

1) The iSAID [6] dataset is derived from multiple remote sensing sensor acquisitions, including 15 types of foreground semantic targets such as airplanes. It provides 655 451 instance annotations and a total of 2806 HSR

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remote sensing images, making it the largest dataset for instance segmentation of HSR remote sensing images. The original images have different dimensions, with size sizes ranging from 8800 × 800 ∼ 4000 × 13 000 pixels. The predefined number of train, validation, and test sets are 1411, 458, and 937 images, respectively. In this work, we use semantic masks for target segmentation and train and validate the model using predefined training set and validation set, respectively.

2) The Vaihingen dataset, supplied by the International Society for Photogrammetry and Remote Sensing (ISPRS), comprises five categories of foreground semantic targets, including buildings. This dataset consists of a total of 33 meticulously labeled HSR remote sensing images. The average size of the raw images is 2494 × 2064 pixels with three spectral bands of red, green and near-infrared. Based on predefined settings and previous work [55], [56], the number of train, validation, and test sets are divided into 15, 1, and 17 images, respectively.

3) The Potsdam dataset, provided by the International ISPRS, encompasses five categories of foreground semantic targets, including buildings. This dataset comprises a total of 38 meticulously labeled HSR remote sensing images. The dimension size of the original images is 6000 × 6000 pixels with four spectral bands of red, green, blue, and near-infrared. Based on the predefined settings and previous work [55], [56], the number of train, validation, and test sets are divided into 22, 1, and 14, respectively, after excluding one mislabeled image. In this work, we use red, green, and blue channel data for experimental validation.

*2) Implementation Detail:* Our method is based on the MMSegmentation implementation and follows the default training settings. Referring to the PFSegNet [8] experimental setup, HRNetv2-w48 [19] is used as the backbone in all our experiments and initialized with ImageNet-1k pretraining parameters. In all experiments, we use a polynomial decay strategy to adjust the learning rate, starting at 0.01. We selected the SGD optimization algorithm for model training, with the weight decay set to 0.0001 and the momentum at 0.99. To enhance the model generalization ability, we use random scale dithering, level flipping, cropping, and color dithering for data preprocessing in the model training process. For dataset-specific configurations, iSAID dataset images are cropped to 896 × 896 pixels using a sliding window of 512 pixels and 80k iterations. For Vaihingen, dataset images are cropped to 768 × 768 pixels using a sliding window of 512 pixels and 40k iterations. For Postdam, dataset images are cropped to 896 × 896 pixels using a sliding window of 512 pixels and 40k iterations. Ablation experiments are conducted over 80k iterations with hyperparameters set to: $m = 0.99$, $\delta = 0.1$, $\lambda = 0.45$, $\mu_1 = 0.01$, $\mu_2 = 0.01$, $\varepsilon = 0.05$, and $l_{\text{start}} = 0.75$. All models were trained on 1 NVIDIA A100 GPU with 80 GB of memory.

*3) Evaluation Metric:* Following conventions [17], [18], we used the mean intersection and merger ratio (mIoU) as an evaluation metric to assess the effectiveness of the method.

TABLE I
COMPARISON WITH THE STATE-OF-THE-ART RESULTS ON iSAID DATASET

| Method | mIoU(%) | #Param (M) | FLOPs(G) |
|---|---|---|---|
| DenseASPP[5] | 57.3 | 27.0 | 353.9 |
| Deeplabv3+[57] | 61.2 | 26.6 | 112.4 |
| RefineNet[58] | 60.2 | 94.9 | 706.9 |
| PSPNet[20] | 60.3 | 46.5 | 546.4 |
| OCNet-(ASP-OC)[59] | 40.2 | 33.8 | 521.9 |
| EMANet[33] | 55.4 | 39.9 | 516.0 |
| CCNet[31] | 58.3 | 47.4 | 613.9 |
| EncodingNet[60] | 58.9 | 33.6 | 501.0 |
| SemanticFPN[23] | 62.1 | 26.1 | 95.8 |
| UPerNet[25] | 63.8 | 35.1 | 235.0 |
| HRNet[19] | 61.8 | 19.2 | 113.2 |
| SFNet[61] | 63.3 | 31.3 | 254.3 |
| GSCNN[62] | 63.4 | 47.2 | 605.6 |
| RANet[63] | 62.1 | 50.1 | 349.4 |
| FarSeg[7] | 63.7 | 31.3 | 144.7 |
| **RSProtoSeg (Ours)** | **65.9** | **71.1** | **273.2** |

TABLE II
COMPARISON WITH THE STATE-OF-THE-ART RESULTS ON VAHIHIGEN (LEFT) AND POTSDAM (RIGHT) DATASETS

| Method | mIoU(%) | mIoU(%) |
|---|---|---|
| DenseASPP[5] | 64.7 | 73.9 |
| Deeplabv3+[57] | 64.3 | 74.1 |
| PSPNet[20] | 65.1 | 73.9 |
| OCNet-(ASP-OC)[59] | 65.7 | 74.2 |
| EMANet[33] | 65.6 | 72.9 |
| CCNet[31] | 64.3 | 73.8 |
| EncodingNet[60] | 65.5 | 73.4 |
| SemanticFPN[23] | 66.3 | 74.3 |
| UPerNet[25] | 66.9 | 74.3 |
| HRNet[19] | 66.9 | 73.4 |
| SFNet[61] | 67.6 | 74.3 |
| GSCNN[62] | 67.7 | 73.4 |
| RANet[63] | 66.1 | 73.8 |
| PFNet[8] | 70.4 | 75.4 |
| **RSProtoSeg (Ours)** | **70.7** | **75.5** |

### B. Comparison to General Methods

*1) Comparison With the State-of-the-Arts on iSAID:* To validate the effectiveness of our methods, comprehensive experiments were conducted on the iSAID dataset. To ensure the fairness of the experiments, ResNet50 served as the backbone for all methods, except those using HRNet. The experiments were conducted under the same setup conditions. The quantitative results of FLOPs from Table I show that compared to previous work [8], this study achieves a better trade-off between model calculation efficiency and accuracy in HSR remote sensing image segmentation scenarios, with high segmentation accuracy at lower computational cost. In addition, our model has 71.1 M parameters, less than 100 M like in prior work, achieving a good balance between fewer parameters and segmentation accuracy. Through optimizing the spatial structure between foreground–background prototypes and intraclass prototypes, we effectively address

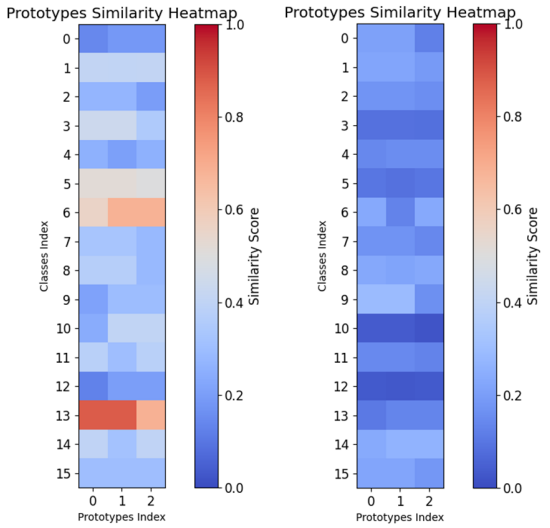
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TABLE III
ABLATIVE STUDIES OF OBJECT SEGMENTATION MIOU (%) ON ISAID VALIDATION SET

| Method | #Prototype | mIoU(%) | # Param (M) |
|---|---|---|---|
| ProtoSegNet[12] | $K$=3 | 64.2 | 71.1 |
| + Online discrete clustering | $K$=3 | 64.8 ↑**0.6** | 71.1 |
| +$L_{FBD}$ | $K$=3 | 65.3 ↑**1.1** | 71.1 |
| + Online discrete clustering +$L_{FBD}$ | $K$=3 | 65.7 ↑**1.5** | 71.1 |

foreground–background imbalance and intraclass variance in HSR remote sensing scenes, outperforming FarSeg by 2.2% in mIoU.

*2) Comparison With the State-of-the-Arts on Vaihingen and Potsdam:* To validate the robustness of our method, scalability experiments are conducted on the Vaihingen and Postdam datasets, respectively. To ensure the fairness of the experiments, the model backbone selection and parameter settings are consistent with the experiments on the iSAID dataset. The quantitative results from Table II show that the mIoU of our method on the two benchmark datasets are70.7and 75.5, achieving state-of-the-art results. It is shown that the issue of foreground–background imbalances and high intraclass variance is common in different HSR remote sensing image datasets. The above problems can be effectively solved by increasing the sparsity between foreground and background prototypes and the diversity of intraclass prototypes, which further verifies the robustness of our method.



(a) (b)

Fig. 5. Intraclass prototype similarity comparison. (a) ProtoSegNet. (b) +Online discrete clustering.

## C. Ablation Study

In this section, we provide a comprehensive analysis of the improvement points and important hyperparameters of the proposed method with the experimental results. This primarily involves analyzing the impact of an online discrete clustering module, foreground–background distance optimization, and the quantity of in-class prototypes on segmentation accuracy, as well as investigating the scalability of lexical semantic segmentation. The mIoU was evaluated on the iSAID validation set with the same experimental settings, if not specified.

*1) Effect of Online Discrete Clustering Module:* Table III shows that our improved online discrete clustering module improves the mIoU by 0.6% compared to ProtoSegNet. Meanwhile, the segmentation accuracy is slightly improved without any parametric increase, indicating that our improved method has better scalability. In addition, it can be seen intuitively from Fig. 5 that the different prototypes in the ProtoSegNet class are dark in color and have high similarity, while the different prototypes in the improved online discrete clustering model are light in color and have low similarity. This shows that our improved online discrete module can effectively reduce the similarity of class prototypes and make different prototypes in the class sparse. Therefore, the quantitative results from Table III and qualitative results from Fig. 5 illustrate that the diversity of intraclass prototypes can effectively solve the issue of large intraclass variance in remote sensing images.

*2) Effect of Foreground–Background Distance Optimization:* Table III presents quantitative results indicating that our enhanced foreground–background distance optimization loss function yields a 1.1% improvement in mIoU value compared to ProtoSegNet. Moreover, this enhancement significantly boosts segmentation accuracy without the need for additional parametric quantities. This suggests that the enhanced foreground–background distance optimization loss function can adaptively adjust the focus on reground weights during the training process, aligning with prototype updates. This effectively addresses the issue of imbalances in the foreground–background distribution. Furthermore, the mIoU of the refined foreground–background distance optimization loss function demonstrates a 0.5% enhancement upon the integration of the improved online discrete clustering module. This underscores the notable impact of addressing foreground–background imbalances in substantially improving the segmentation accuracy of HSR remote sensing images.

TABLE IV
ABLATIVE STUDIES WITH PROTOTYPE NUMBER PER CLASS $K$ OF OBJECT SEGMENTATION MIOU (%) ON ISAID VALIDATION SET

| #Prototype | mIoU(%) | # Param (M) |
|---|---|---|
| $K$=1 | 65.1 | 71.1 |
| $K$=3 | 65.7 | 71.1 |
| $K$=5 | 65.4 | 71.1 |
| $K$=10 | 65.9 | 71.1 |

*3) Performance on Segmentation:* Table III Quantitative results show that the mIoU is improved by 1.5% compared to ProtoSegNet after we combine the improved online

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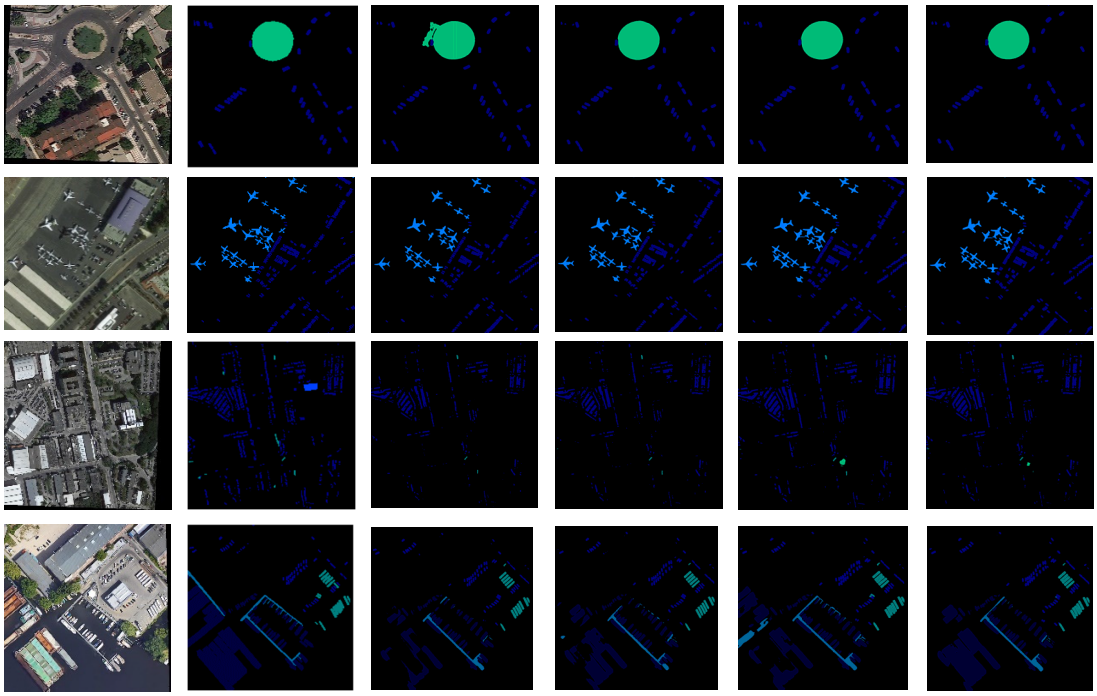



Fig. 6. Visual results on iSAID validation set. Online discrete clustering module and foreground–background distance optimization. Best view on the screen.

TABLE V
WE STUDY THE SCALABILITY OF THE PROPOSED METHOD ON POSTDAM AND ISAID DATASETS

| Method | #Prototype | 6 classes (Postdam) | | 16 classes (iSAID) | |
|---|---|---|---|---|---|
| | | # Param (M) | mIoU(%) | #Param(M) | mIoU(%) |
| RSProtoSeg (Ours) | $K$=3 | 71.1 | 75.1 | 71.1 | 65.7 |
| | $K$=10 | 71.1 | 75.5 | 71.1 | 65.9 |

discrete clustering module and foreground–background distance optimized loss function. Meanwhile, the segmentation accuracy is improved substantially without adding any parametric quantities. Additionally, when compared with the individual addition of either the online discrete clustering module or the foreground–background distance optimization loss function, our proposed method yields improvements of 0.9% and 0.4% in mIoU values, respectively. This underscores the effectiveness of our approach in optimizing both the interstructure between foreground and background prototypes and the spatial structure of prototypes within the class. Consequently, it addresses issues related to imbalanced foreground–background distribution and large intraclass variance.

*4) Visual Results Comparison:* In Fig. 6, we qualitatively explore the effects of online discrete clustering module and foreground–background distance optimization on segmentation accuracy. Our proposed online discrete clustering module has better segmentation results when dealing with intraclass difference variance semantics. In addition, our proposed foreground–background distance optimization has clear target boundaries and better segmentation results in dealing with foreground false positives.

*5) Prototype Number per Class K:* Table IV shows the segmentation performance of our proposed method for different number of prototypes within a class. When the number of prototypes within the class is set to 1, we adopt the average pixel features within the class as the feature representation for each semantic category. In this scenario, classification into pixels relies solely on the truth labels, resulting in mIoU value of 65.1%. Furthermore, as the number of intraclass prototypes increases (e.g., $K = 3$), the segmentation performance is significantly improved (e.g., 65.1% $\rightarrow$ 65.7%). When continuing to increase the number of in-class prototypes, the segmentation performance tends to fluctuate and rebound, but remains higher than when the number of in-class prototypes is 1. In particular, despite the significant improvement in segmentation performance as the number of in-class prototypes increases, there is no increase in any number of parameters. This study verifies that multiple prototypes within classes can solve the issue of large intraclass variance.

*6) Scalability to Vocabulary Semantic Segmentation:* Table V Quantitative results demonstrate two key findings. First, our proposed method maintains a consistent number of parameters even as the number of semantic categories increases. Second, it also does not increase the number

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of parameters as the number of prototypes within a class increases. This study verifies that our proposed method is not limited by the number of semantic categories and is highly scalable.

## V. Conclusion

In this study, we introduce a remote sensing image segmentation algorithm based on the nonparametric segmentation paradigm of unlearnable prototypes. The aim is to address the challenges posed by unbalanced foreground–background distributions and significant intraclass variance in the segmentation of HSR remote sensing images. The algorithm is explicitly optimized from the perspective of the spatial structure between foreground and background prototypes as well as intraclass prototypes. The design of the foreground–background distance optimization loss function and the online discrete clustering module are designed respectively to make it suitable for HSR remote sensing image segmentation. Comprehensive experimental results show that our proposed method achieves advanced segmentation results with high scalability without increasing any number of parameters. Further experiments on three generalized segmentation datasets also demonstrate the generality of the method.

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## References

[1] M. Volpi and V. Ferrari, "Semantic segmentation of urban scenes by learning local class interactions," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2015, pp. 1–9.

[2] R. Kemker, C. Salvaggio, and C. Kanan, "Algorithms for semantic segmentation of multispectral remote sensing imagery using deep learning," *ISPRS J. Photogramm. Remote Sens.*, vol. 145, pp. 60–77, Nov. 2018.

[3] D. He, Y. Zhong, and L. Zhang, "Spectral–spatial–temporal MAP-based sub-pixel mapping for land-cover change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 3, pp. 1696–1717, Mar. 2020.

[4] G.-S. Xia et al., "DOTA: A large-scale dataset for object detection in aerial images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 3974–3983.

[5] M. Yang, K. Yu, C. Zhang, Z. Li, and K. Yang, "DenseASPP for semantic segmentation in street scenes," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 3684–3692.

[6] S. W. Zamir et al., "iSAID: A large-scale dataset for instance segmentation in aerial images," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. Workshops. (CVPRW)*, Jun. 2019, pp. 28–37.

[7] Z. Zheng, Y. Zhong, J. Wang, and A. Ma, "Foreground-aware relation network for geospatial object segmentation in high spatial resolution remote sensing imagery," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 4095–4104.

[8] X. Li et al., "PointFlow: Flowing semantics through points for aerial image segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 4215–4224.

[9] F. Yang and C. Ma, "Sparse and complete latent organization for geospatial semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 1799–1808.

[10] H. Zhang, K. Zu, J. Lu, Y. Zou, and D. Meng, "EPSANet: An efficient pyramid squeeze attention block on convolutional neural network," in *Proc. Asia. Conf. Comput. Vis. (ACCV)*, Jul. 2022, pp. 1161–1177.

[11] H. Fan et al., "Multiscale vision transformers," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 6804–6815.

[12] T. Zhou, W. Wang, E. Konukoglu, and L. Van Goo, "Rethinking semantic segmentation: A prototype view," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 2572–2583.

[13] N. Otsu, "A threshold selection method from gray-level histograms," *IEEE Trans. Syst., Man, Cybern.*, vol. SMC-9, no. 1, pp. 62–66, Jan. 1979.

[14] R. Nock and F. Nielsen, "Statistical region merging," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 26, no. 11, pp. 1452–1458, Nov. 2004.

[15] N. Plath, M. Toussaint, and S. Nakajima, "Multi-class image segmentation using conditional random fields and global classification," in *Proc. 26th Annu. Int. Conf. Mach. Learn.*, Jun. 2009, pp. 817–824.

[16] J.-L. Starck, M. Elad, and D. L. Donoho, "Image decomposition via the combination of sparse representations and a variational approach," *IEEE Trans. Image Process.*, vol. 14, no. 10, pp. 1570–1582, Oct. 2005.

[17] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 3431–3440.

[18] L.-C. Chen, G. Papandreou, I. Kokkinos, K. Murphy, and A. L. Yuille, "DeepLab: Semantic image segmentation with deep convolutional nets, atrous convolution, and fully connected CRFs," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 4, pp. 834–848, Apr. 2018.

[19] J. Wang et al., "Deep high-resolution representation learning for visual recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 10, pp. 3349–3364, Oct. 2021.

[20] H. Zhao, J. Shi, X. Qi, X. Wang, and J. Jia, "Pyramid scene parsing network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 6230–6239.

[21] W. Chen, Z. Jiang, Z. Wang, K. Cui, and X. Qian, "Collaborative global-local networks for memory-efficient segmentation of ultra-high resolution images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 8916–8925.

[22] P. Bilinski and V. Prisacariu, "Dense decoder shortcut connections for single-pass semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 6596–6605.

[23] A. Kirillov, R. Girshick, K. He, and P. Dollár, "Panoptic feature pyramid networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 6392–6401.

[24] X. Li, H. Zhao, L. Han, Y. Tong, S. Tan, and K. Yang, "Gated fully fusion for semantic segmentation," in *Proc. AAAI Conf. Artif. Intell.*, Feb. 2020, no. 7, pp. 11418–11425.

[25] T. Xiao, Y. Liu, B. Zhou, Y. Jiang, and J. Sun, "Unified perceptual parsing for scene understanding," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Sep. 2018, pp. 418–434.

[26] Z. Zhang, X. Zhang, C. Peng, X. Xue, and J. Sun, "ExFuse: Enhancing feature fusion for semantic segmentation," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Sep. 2018, pp. 269–284.

[27] E. Xie et al., "SegFormer: Simple and efficient design for semantic segmentation with transformers," in *Proc. NIPS*, 2021, pp. 12077–12090.

[28] W. Wang et al., "Pyramid vision transformer: A versatile backbone for dense prediction without convolutions," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 568–578.

[29] B. Cheng, A. G. Schwing, and A. Kirillov, "Per-pixel classification is not all you need for semantic segmentation," in *Proc. NIPS*, Dec. 2021, pp. 17864–17875.

[30] S. Ren, D. Zhou, S. He, J. Feng, and X. Wang, "Shunted self-attention via multi-scale token aggregation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 10843–10852.

[31] Z. Huang, X. Wang, L. Huang, C. Huang, Y. Wei, and W. Liu, "CCNet: Criss-cross attention for semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 603–612.

[32] A. Tao, K. Sapra, and B. Catanzaro, "Hierarchical multi-scale attention for semantic segmentation," 2020, *arXiv:2005.10821*.

[33] X. Li, Z. Zhong, J. Wu, Y. Yang, Z. Lin, and H. Liu, "Expectation-maximization attention networks for semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 9166–9175.

[34] L. Ding, H. Tang, and L. Bruzzone, "LANet: Local attention embedding to improve the semantic segmentation of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 426–435, Jan. 2021.

[35] L. Mou, Y. Hua, and X. X. Zhu, "Relation matters: Relational context-aware fully convolutional network for semantic segmentation of high-resolution aerial images," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 11, pp. 7557–7569, Nov. 2020.

[36] Y. Liu, Y. Zhang, Y. Wang, and S. Mei, "Rethinking transformers for semantic segmentation of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5617515.

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[37] Y. Liu, Y. Zhang, Y. Wang, and S. Mei, "BiTSRS: A bi-decoder transformer segmentor for high-spatial-resolution remote sensing images," *Remote Sens.*, vol. 15, no. 3, p. 840, Feb. 2023.

[38] B. Huang, B. Zhao, and Y. Song, "Urban land-use mapping using a deep convolutional neural network with high spatial resolution multispectral remote sensing imagery," *Remote Sens. Environ.*, vol. 214, pp. 73–86, Sep. 2018.

[39] C. Zhang et al., "Joint deep learning for land cover and land use classification," *Remote Sens. Environ.*, vol. 221, pp. 173–187, Feb. 2019.

[40] J. Yuan, "Learning building extraction in aerial scenes with convolutional networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 11, pp. 2793–2798, Nov. 2018.

[41] J. Gawlikowski, S. Saha, A. Kruspe, and X. X. Zhu, "An advanced Dirichlet prior network for out-of-distribution detection in remote sensing," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5616819.

[42] W. Chen, G. Zhou, Z. Liu, X. Li, X. Zheng, and L. Wang, "NIGAN: A framework for mountain road extraction integrating remote sensing road-scene neighborhood probability enhancements and improved conditional generative adversarial network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5626115.

[43] J. Liang, N. Homayounfar, W.-C. Ma, S. Wang, and R. Urtasun, "Convolutional recurrent network for road boundary extraction," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 9504–9513.

[44] J. Yang, J. Guo, H. Yue, Z. Liu, H. Hu, and K. Li, "CDnet: CNN-based cloud detection for remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 6195–6211, Aug. 2019.

[45] C. Lu, M. Xia, M. Qian, and B. Chen, "Dual-branch network for cloud and cloud shadow segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5410012.

[46] C. Decaestecker, "Finding prototypes for nearest neighbour classification by means of gradient descent and deterministic annealing," *Pattern Recognit.*, vol. 30, no. 2, pp. 281–288, Feb. 1997.

[47] T. Kohonen, "The self-organizing map," *Proc. IEEE*, vol. 78, no. 9, pp. 1464–1480, Sep. 1990.

[48] A. Sato and K. Yamada, "Generalized learning vector quantization," in *Proc. Adv. Neural Inf. Process. Sys. (NIPS)*, vol. 8, 1998, pp. 423–429.

[49] H.-M. Yang, X.-Y. Zhang, F. Yin, and C.-L. Liu, "Robust classification with convolutional prototype learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 3474–3482.

[50] G. Li, V. Jampani, L. Sevilla-Lara, D. Sun, J. Kim, and J. Kim, "Adaptive prototype learning and allocation for few-shot segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 8330–8339.

[51] Y. Yu, Z. Ji, J. Han, and Z. Zhang, "Episode-based prototype generating network for zero-shot learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 14032–14041.

[52] T. Monnier, E. Vincent, J. Ponce, and M. Aubry, "Unsupervised layered image decomposition into object prototypes," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 8620–8630.

[53] M. Nauta, R. van Bree, and C. Seifert, "Neural prototype trees for interpretable fine-grained image recognition," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 14928–14938.

[54] M. Cuturi, "Sinkhorn distances: Lightspeed computation of optimal transport," in *Proc. Adv. Neural Inf. Process. Sys. (NIPS)*, vol. 26, Dec. 2013, pp. 2292–2301.

[55] R. Li et al., "Multiattention network for semantic segmentation of fine-resolution remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2021, Art. no. 5607713.

[56] X. Qi, Y. Mao, Y. Zhang, Y. Deng, H. Wei, and L. Wang, "PICS: Paradigms integration and contrastive selection for semisupervised remote sensing images semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5602119.

[57] L.-C. Chen, Y. Zhu, G. Papandreou, F. Schroff, and H. Adam, "Encoder–decoder with atrous separable convolution for semantic image segmentation," in *Proc. Eur. Conf Comput. Vis. (ECCV)*, Sep. 2018, pp. 801–818.

[58] G. Lin, A. Milan, C. Shen, and I. Reid, "RefineNet: Multi-path refinement networks for high-resolution semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 5168–5177.

[59] Y. Yuan, L. Huang, J. Guo, C. Zhang, X. Chen, and J. Wang, "OCNet: Object context network for scene parsing," 2018, *arXiv:1809.00916*.

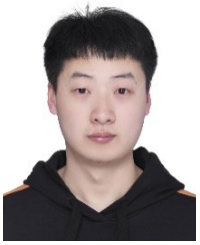
[60] H. Zhang et al., "Context encoding for semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7151–7160.

[61] X. Li et al., "Semantic flow for fast and accurate scene parsing," in *Proc. Eur. Conf Comput. Vis. (ECCV)*, 2020, pp. 775–793.

[62] T. Takikawa, D. Acuna, V. Jampani, and S. Fidler, "Gated-SCNN: Gated shape CNNs for semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 5228–5237.

[63] L. Mou, Y. Hua, and X. X. Zhu, "A relation-augmented fully convolutional network for semantic segmentation in aerial scenes," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 12408–12417.

[64] M. Berman, A. R. Triki, and M. B. Blaschko, "The Lovász-softmax loss: A tractable surrogate for the optimization of the intersection-over-union measure in neural networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 4413–4421.



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